

## Project Design Phase-I

### Solution Architecture

|               |                                                  |
|---------------|--------------------------------------------------|
| Date          | 9 March 2026                                     |
| Team ID       | SWUID202641986                                   |
| Project Name  | Insurance Fraud Detection Using Machine Learning |
| Maximum Marks | 4 Marks                                          |

### Solution Architecture: Insurance Fraud Detection System

The Insurance Fraud Detection System is a machine learning-powered web application designed to help insurance companies identify potentially fraudulent automobile insurance claims. It bridges the gap between raw insurance data and actionable fraud insights.

#### System Overview

The system follows a classic 3-tier architecture with a decoupled Machine Learning engine.

#### 1. User Interface (Frontend)

- Technology: HTML5, Vanilla CSS, JavaScript.
- Function: Provides a responsive web form for insurance agents to input claim details (age, annual premium, total claim amount, etc.) and displays prediction results with probability scores.

#### 2. Application Server (Backend)

- Technology: Python, Flask.
- Function: Orchestrates data flow between the UI and the ML model. It handles HTTP requests, validates input data, and triggers the ML pipeline.

#### 3. Machine Learning Engine

- Preprocessing: StandardScaler (via scaler.pkl) ensures input data matches the training distribution.
- Model: Random Forest / Gradient Boosting (via model.pkl) performs binary classification (Fraud vs. Genuine).
- Output: Predicts the likelihood of fraud as a percentage.

## Solution Architecture Diagram:

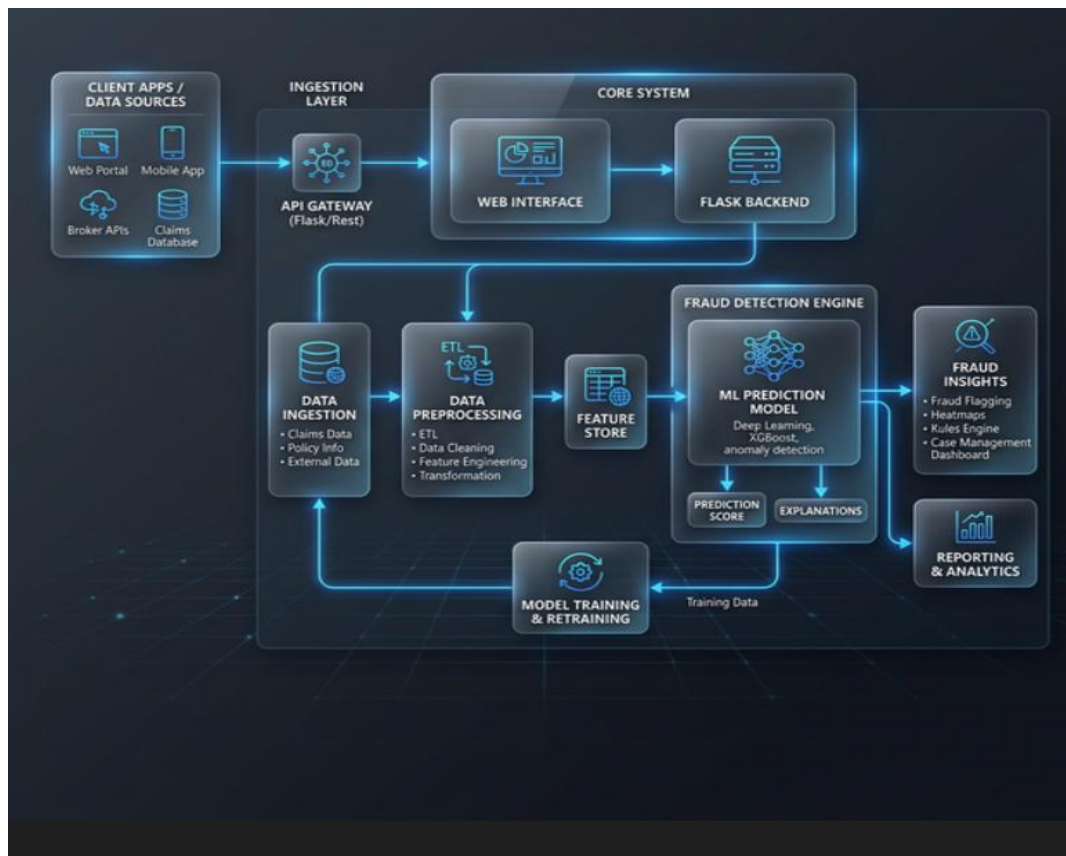


Figure 1: Architecture and data flow of the Insurance Fraud Detection Using Machine Learning

### Reference:

- **AWS Reference:** [Architecture and design considerations for voice applications](#) (The source you mentioned)
- **Microsoft Azure:** [Solution architecture design principles](#)
- **Google Cloud:** [Architecture Framework & Best Practices](#)
- **IBM:** [Solution Architecture Overview](#)